# Software Test Plan (STP)

## Online Bookstore

**Version 1.0**

| Item | Details |
|---|---|
| Project | Online Bookstore |
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| Date | 16-09-2026 |
| Status | Draft |
| Reference Documents | Online Bookstore SRS v1.0 (08-09-2026)<br>Online Bookstore SAD v1.1 (12-09-2026) |

## 1. Introduction

This Software Test Plan (STP) defines the testing objectives, scope, strategy, resources, schedule, responsibilities, risks, traceability, and reporting approach for Version 1.0 of the Online Bookstore. It is derived from the team's SRS and SAD.

Scope: Testing covers customer account management, catalogue/search, shopping cart, checkout and delivery information, payment processing, order management and tracking, inventory management, administrator operations, security controls, and specified non-functional requirements. Physical warehouse operations, book publishing, physical delivery operations, banking infrastructure, and internal implementation of third-party payment/courier services are outside scope.

References: Online Bookstore SRS v1.0; Online Bookstore SAD v1.1; Online Bookstore Project Proposal; PES University Software Engineering course materials; IEEE Std 830-1998.

Definitions: SRS (Software Requirements Specification), SAD (Software Architecture and Design Specification), STP (Software Test Plan), RTM (Requirements Traceability Matrix), TC (Test Case), RBAC (Role-Based Access Control), API (Application Programming Interface), DBMS (Database Management System), NFR (Non-Functional Requirement).

## 2. Test Items

- Customer registration, login, authentication, authorization, and profile management.
- Book catalogue browsing, search, filtering, sorting, and book-detail display.
- Shopping cart add, quantity update, and remove operations.
- Checkout, delivery information, and order summary.
- Payment processing and payment-result verification through the payment gateway/API.
- Order creation, unique Order ID generation, order viewing, tracking, and eligible cancellation.
- Inventory quantity management and stock enforcement during checkout.
- Administrator book, pricing, category, inventory, and order management.
- Security controls: password protection, RBAC, HTTPS/TLS, session security, transaction integrity, and payment-data protection.
- Performance, scalability, reliability, usability, maintainability, and portability.

# 3. Features to be Tested

The SRS does not define individual FR identifiers. Therefore, this test plan uses SRS section numbers and the explicit NFR/security identifiers for traceability, consistent with the SAD.

| Requirement / Feature | Planned validation |
|---|---|
| SRS 5.1 – Customer Account Management | Registration, login, authentication, authorization, profile management |
| SRS 5.2 – Book Catalogue and Search | Browse, title/author/category/ISBN search, filtering, sorting, details |
| SRS 5.3 – Shopping Cart | Add, quantity update, remove items |
| SRS 5.4 – Checkout and Delivery | Cart review, delivery details, order summary |
| SRS 5.5 – Payment Processing | Payment request, response validation, failure and duplicate prevention |
| SRS 5.6 – Order Management | Order ID, history, tracking, eligible cancellation |
| SRS 5.7 / BR-1 – Inventory Management | Stock updates and insufficient-stock prevention |
| SRS 5.8 / SEC-8 – Administrator Management | Authorized CRUD, inventory and order management |
| SRS 5.9 / BR-2 – Order Tracking and Status | Status display and permitted status handling |
| SRS 5.10 / SEC-1 to SEC-8 – Security | Authentication, hashing, RBAC, TLS, sessions, transaction/payment protection |
| PERF-1 to PERF-4 | Response time, checkout, 500 concurrent users, page load |
| SCAL-1 to SCAL-3 | 100,000-title catalogue and application/database scaling |
| REL-1 to REL-4 | 99% uptime, failure handling, backups, recovery |
| USE-1 to USE-4 | Navigation, messages, consistency, administrator learnability |
| MAIN-1 to MAIN-3 | Modularity, logging, documentation |
| PORT-1 to PORT-2 | Browser and desktop/tablet/mobile compatibility |

# 4. Features Not to be Tested

- Physical warehouse operations.
- Book publishing activities.
- Physical delivery/courier operations and internal courier implementation.
- Banking infrastructure and internal banking-system implementation.
- Internal implementation of the third-party payment gateway; only the bookstore's integration and failure handling are tested.
- Third-party infrastructure not controlled by the project team.

# 5. Test Approach / Strategy

## 5.1 Test Levels

- Unit Testing – Individual modules such as authentication, catalogue/search, cart, checkout, inventory, and order logic.
- Integration Testing – Interactions among React UI, Node.js/Express REST API, MySQL, and payment gateway/API.
- System Testing – Complete end-to-end customer and administrator workflows.
- Acceptance Testing (UAT) – Verification against SRS requirements and usability expectations.

### 5.2 Test Types

- Functional testing – Core customer, administrator, payment, order, and inventory behaviour.
- Negative and boundary testing – Invalid credentials/input, empty cart, excessive quantity, unavailable stock, invalid payment responses, and unauthorized access.
- Regression testing – Re-run affected tests after defect fixes or changes.
- Performance testing – Response time, page load, checkout processing, and concurrent-user behaviour.
- Security validation – Authentication, RBAC, password protection, TLS, session security, transaction integrity, and payment-data handling.
- Usability and compatibility testing – Navigation, messages, consistency, learnability, browsers, and responsive layouts.
- Recovery testing – Third-party/payment failure, backup restoration, and transaction consistency.

### 5.3 Entry Criteria

- Stable testable build is available.
- Required database schema and representative test data are available.
- Test environment and required APIs are accessible.
- Major blocking defects are resolved or formally accepted.
- Test cases and expected results are reviewed.

### 5.4 Exit Criteria

- 100% of planned test cases have been executed for the release scope.
- No unresolved Critical defects remain.
- All planned SRS requirements have corresponding test coverage.
- Required NFR/security tests have been executed and recorded.
- UAT acceptance criteria are satisfied or formally waived.

### 5.5 Security Validation

- Verify authentication before protected functions and administrator operations.
- Verify passwords are not stored in plain text and use secure hashing.
- Verify RBAC prevents customers from accessing administrator operations.
- Verify HTTPS/TLS for sensitive client/server and payment communication.
- Verify authenticated sessions are invalidated on logout.
- Verify input validation and transaction consistency prevent duplicate/inconsistent orders.
- Verify raw payment credentials are not unnecessarily stored.
- Verify significant errors and administrative actions are logged without exposing sensitive customer data.

## 6. Test Environment

| Area | Environment / Configuration |
|---|---|
| Client hardware | Desktop/laptop plus representative tablet and smartphone screen sizes |
| Client software | Latest stable Chrome, Firefox, Safari, and Edge |
| Frontend | React.js web interface |
| Backend | Node.js + Express.js REST API |
| Database | MySQL |
| External service | Payment Gateway/API sandbox or controlled test environment |
| Network | Stable internet connection; HTTPS/TLS enabled |
| API testing | Postman or equivalent (proposed) |

| | |
|---|---|
| UI automation | Selenium or equivalent (proposed) |
| Performance | JMeter or equivalent (proposed) |
| Defect tracking | Jira/GitHub Issues or equivalent (proposed) |
| Test data | Dummy customer/admin accounts, books, categories, inventory, carts, orders, and payment responses |

Note: The SRS/SAD specify the application technology and logical interfaces but do not mandate particular testing tools. The tools above are proposed choices for the test team.

## 7. Test Schedule

The following is a proposed schedule for the assignment and may be adjusted to the implementation timeline.

| Milestone | Proposed Date |
|---|---|
| Test plan and test-case design | 17-Sep-2026 |
| Test environment setup | 18-Sep-2026 |
| Unit and integration testing | 19-Sep-2026 to 21-Sep-2026 |
| System and functional testing | 22-Sep-2026 to 25-Sep-2026 |
| Security / performance / compatibility testing | 26-Sep-2026 to 27-Sep-2026 |
| Defect retesting and regression | 28-Sep-2026 to 29-Sep-2026 |
| UAT | 30-Sep-2026 to 01-Oct-2026 |
| Final test summary and sign-off | 02-Oct-2026 |

## 8. Test Deliverables

- Software Test Plan (STP).
- Test case specification and execution records.
- Manual and automated test scripts, where applicable.
- Test data and test-environment configuration.
- API/UI/performance test results.
- Defect reports and retest evidence.
- Requirements Traceability Matrix (RTM).
- Final Test Summary Report.

## 9. Roles and Responsibilities

| Role | Name | Responsibility |
|---|---|---|
| QA Lead | Varchas | Prepare/maintain STP, coordinate execution, review coverage and results. |
| Test Engineer | Uditi Mourya | Design/execute tests, perform regression, record defects. |
| Developer / Test Support | Vrinda Venugopal M | Support builds, defect diagnosis/fixes, and technical testing. |
| Product Owner / Evaluator | Vrinda Venugopal M | Review acceptance criteria, test results, and provide sign-off. |

## 10. Risks and Mitigation

| Risk | Mitigation |
|---|---|

| | |
|---|---|
| Delay in stable build delivery | Request early smoke builds and prepare test cases before full build availability. |
| Test environment/database downtime | Maintain reproducible configuration and backups; use a backup environment where feasible. |
| Payment gateway/API unavailable | Use sandbox/mock responses and test failure/rollback handling. |
| Insufficient test data | Prepare representative customer, book, inventory, order, and payment datasets. |
| Duplicate order/payment requests | Test transaction consistency and verify failed requests do not create duplicates. |
| Insufficient stock under concurrent checkout | Execute concurrent purchase tests and verify stock/order consistency. |
| Unauthorized administrative access | Perform RBAC and authorization tests for customer and administrator roles. |
| Cross-browser/responsive defects | Test specified browsers and desktop/tablet/mobile layouts. |

## 11. Assumptions & Dependencies

- React.js frontend, Node.js/Express.js backend, and MySQL database are available for testing.
- Representative test data and test accounts can be created without using real customer/payment credentials.
- A payment gateway/API sandbox or controlled mock response is available for integration testing.
- External delivery tracking is treated as an external dependency; only specified internal handling is tested.
- HTTPS/TLS configuration is available in the test environment.
- Testing is performed on supported modern browsers and representative device sizes.
- Proposed testing tools are available if automation/performance testing is required.

## 12. Suspension & Resumption Criteria

Suspend testing if:

- Test environment or a critical dependency is unavailable for more than 4 hours.
- The build is sufficiently unstable that more than 30% of planned tests are blocked.
- A Critical defect prevents meaningful execution of major purchase, authentication, or administrator flows.
- Database/test data becomes unreliable or corrupted.

Resume testing when:

- Blocking/Critical defects are resolved or formally accepted.
- Environment and dependencies are stable.
- Required test data is restored or recreated.
- A smoke test confirms that the build is suitable for continued execution.

## 13. Test Case Management & Traceability

The RTM maps each SRS requirement or requirement group to one or more test cases. Since the SRS does not define individual FR identifiers, traceability uses SRS section numbers and explicit NFR/security identifiers.

| Traceability Item | Example Test Cases |
|---|---|
| SRS 5.1 – Customer Account Management | TC-AUTH-01 valid registration; TC-AUTH-02 invalid login; TC-AUTH-03 protected-page authorization |
| SRS 5.2 – Catalogue/Search | TC-CAT-01 title search; TC-CAT-02 author/category/ISBN search; TC-CAT-03 filtering/sorting |
| SRS 5.3 – Shopping Cart | TC-CART-01 add item; TC-CART-02 quantity update; TC-CART-03 remove item |

| | |
|---|---|
| SRS 5.4 – Checkout | TC-CHK-01 delivery details; TC-CHK-02 order summary; TC-CHK-03 invalid checkout input |
| SRS 5.5 / SEC-7 – Payment | TC-PAY-01 successful payment; TC-PAY-02 failed payment; TC-PAY-03 duplicate request |
| SRS 5.6 / BR-2 – Orders | TC-ORD-01 order creation/ID; TC-ORD-02 history; TC-ORD-03 tracking/cancellation |
| SRS 5.7 / BR-1 – Inventory | TC-INV-01 stock decrement; TC-INV-02 insufficient stock; TC-INV-03 concurrent consistency |
| SRS 5.8 / SEC-8 – Administrator | TC-ADM-01 admin login; TC-ADM-02 book CRUD; TC-ADM-03 inventory/order update |
| PERF-1 to PERF-4 | TC-PERF-01 catalogue response; TC-PERF-02 checkout; TC-PERF-03 500-user load; TC-PERF-04 page load |
| SCAL-1 to SCAL-3 | TC-SCAL-01 100,000-title catalogue; TC-SCAL-02 horizontal app scaling; TC-SCAL-03 database growth |
| REL-1 to REL-4 | TC-REL-01 uptime; TC-REL-02 payment failure rollback; TC-REL-03 backup; TC-REL-04 recovery |
| USE-1 to USE-4 / PORT-1 to PORT-2 | TC-UX-01 three-click navigation; TC-UX-02 error messages; TC-COMP-01 browser/device compatibility |

## 14. Test Metrics & Reporting

Metrics collected:

- Percentage of planned test cases executed.
- Pass/fail/blocked/not-run percentages.
- Requirement coverage percentage.
- Defect count by severity and priority.
- Defect density and defect aging.
- Regression pass rate.
- Performance measurements against PERF-1 to PERF-4.
- Availability/recovery results against REL-1 to REL-4.

Reports:

- Daily or per-cycle test execution status.
- Defect summary and retest status.
- Requirements Traceability Matrix.
- Final Test Summary Report with execution statistics, defects, requirement coverage, NFR/security results, and UAT status.

## 15. Approvals

| Role | Name | Date |
|---|---|---|
| QA Lead | Varchas | 16/09/2026 |
| Test Engineer | Uditi Mourya | 16/09/2026 |
| Developer / Test Support /Product Owner / Evaluator | Vrinda Venugopal M | 16/09/2026 |